Feedback

CBT Tech Climate Solutions Challenge

GSSA - Gravataí Support & Shelter Assist

Team Member 1 Age

> 17

Team Member 1 Country

> Brazil

What interests you most about this challenge?

> I believe that the World Series of Innovation fosters not only student engagement with dilemmas of sudden global relevance, but also provides an opportunity for action and reflection on the adversities that surround us, contemplating how the innovative nature of each individual can contribute to providing a higher quality of life for all. Through initiatives like this, it's possible to consider the existing demands across the globe, how they connect multinationally, how they impact general well-being, and how we can unite in the face of such barriers. I believe that creativity, innovation, and genuine dedication can lead us to extraordinary problem-solving paths. I observe my surroundings daily, thinking about how people, groups, and institutions can have their lives made easier through unique ideas... and I believe that the World Series of Innovation mitigates and propagates opportunities for developing the entrepreneurial spirit and mindset. So many ideas... so many places... so many perspectives. I believe that, in the end, this is what captivates me most in challenges like these: the appreciation of out-of-the-box thinking.

Who is experiencing the problem you are working to solve?

> The dilemma I'm addressing affects three main interconnected groups within climate-vulnerable communities, particularly in peripheral urban areas such as Gravataí, Rio Grande do Sul, Brazil – my hometown, which suffered the worst climate catastrophe in the history not only of southern Brazil, but of the entire country. May 2024's floods won't be forgotten.
>
> The first group includes families and individuals directly exposed to the climate crisis. In other words, people living in areas with fragile infrastructure, insufficient drainage, and limited emergency support. In

Gravataí, thousands of low-income residents, elderly individuals living alone, single moms, and people with disabilities were forced from their homes. In nearby São Leopoldo, nearly half the city's population had to evacuate as more than thirty thousand houses were affected by the 2024 floods. This resulted in 15,000 people needing shelter, without – perhaps – ever having questioned such a need... all in a single city. The impact was statewide. 2,398,255 people in a state of 10 million. All directly impacted.

I saw personally how these people made their way through flooded streets, their goods in backpacks, their children sobbing, and unsure of where to seek refuge. Where to find hope. They knocked on church doors only to discover shelters were full, or walked kilometers in the rain searching for help. What struck me was their isolation. They had no centralized way to discover which institutions were providing assistance, where beds or food were available, or how far they'd need to travel. Many of those individuals own basic smartphones but lack access to organized, real-time information during emergencies.

Furthermore, community centers and non-governmental organizations are the second group involved in this predicament. In peripheral communities, local NGOs, churches, and grassroots groups don't just assist; they essentially serve as the foundation for the entire disaster response. During the 2024 floods, I spoke with administrators from organizations in Gravataí who expressed profound frustration. "We had 40 empty beds while families slept under bridges 'cause they didn't know we existed." Even when my school offered itself as a shelter for those impacted by the floods, it took days for a bus with displaced people from Cachoeirinha to meet us and for them to be fully assisted. Animals arrived in states of extreme immunological fragility, suffering from scratches, injuries, and illnesses requiring treatment. If they'd come quicker, the assistance would have been more efficient. These institutions operate with minimal staff, limited technological capacity, and urgent need for better communication infrastructure.

In closing, the third group impacted by this climatic dilemma is community members wanting to help. During the floods, I observed an extraordinary wave of solidarity. High school students, healthcare professionals, truck drivers, teachers... thousands wanted to contribute but couldn't find verified and organized opportunities. They scrolled through chaotic WhatsApp groups with hundreds of incompatible and disconnected messages, searched Google for outdated information, and often ended up not helping at all simply because reliable channels didn't exist. The willingness to help was immense; however, the infrastructure to channel and transmit that goodwill was absent.

Does the student clearly identify a specific customer or community?

The student clearly identifies three interconnected but specific customer groups, displaced residents in climate vulnerable communities, local NGOs and community institutions, and volunteers seeking to help. The geographic anchoring in Gravataí and nearby cities adds precision and credibility rather than vagueness.

9 months ago

How well does the student understand the customer?

Very strong, personal story helps to define the customer / community well.

9 months ago

The student demonstrates an exceptionally strong understanding of customer context. They clearly show lived experience, empathy, and firsthand observation from the 2024 floods in Rio Grande do Sul. The narrative captures emotional, logistical, and informational pain points across multiple stakeholders, with concrete examples that go well beyond surface level insight.

9 months ago

What problem or need are you trying to solve for your customer?

The main dilemma I seek to address is the imminent collapse of communication during climate emergencies. A collapse that is neither abstract, nor technical, nor distant. It's deeply human. In May 2024, when streets became rivers, I realized that the greatest enemy was not just the water, but the silence between those who needed help and those who could offer it. When communication fails, everything fails. And entire communities pay the price.

The invisibility of resources turns out as the first symptom of this collapse. During the flood, shelters appeared everyplace... churches, schools, community centers, and homes of empathetic families. There were hot meals, dry clothes, and medical care. But for thousands of people, none of this existed – not because it was far away, because it was impossible to see. Elderly people, migrants, and residents of peripheral neighborhoods were, indeed, particularly isolated. The real need is to make visible what already exists, transforming the desperate search for help into immediate guidance. The necessity of turning kindness into action.

The second symptom is the lack of coordination between organizations and potential volunteers. The flood, surely, exposed a distressing paradox: exhausted NGOs begging for support while volunteers wandered around unable to help. Organizations needed specific skills (medical professionals, cooks, truck drivers, translators...) but had no structured to broadcast these demands or verify volunteers's capabilities. Therefore, the underlying need is a methodic matching system where institutions can specify exactly what kind of support they need ("2 volunteers with cooking experience, tomorrow 6pm, location X") and potential volunteers can register with verified skills, availability, and location. This ends up transforming chaotic goodwill into organized, effective action.

Finally, the third symptom shows up even before the crisis: the lack of integrated preparation. The authorities' alerts arrived in April 2024, but they did not reach individuals in an understandable way. Many did not understand what a high-risk warning meant. Nobody expects a climate-crisis. Nobody expects a flood. They didn' t know when to leave. Not where to go. Not even what to protect. Documents, medications, and memories got lost. They evacuated too late or too early. The information existed, but it was not accessible. Preparation cannot be a privilege, it needs to be an everyday knowledge, translated into simple and immediate actions.

The consequences of these failures are devastating. Not only in material or economic losses, but in emotional wear and tear. Each minute of confusion costs security. Each conflicting piece of information

costs dignity. Each silence costs lives.

And this is not just a problem for one city. The flood of 2024 was a Brazilian (and global) mirror. Medium-sized cities, marked by rapid and unequal urbanization, repeat the same pattern... limited institutional capacity, vulnerable populations, and a huge information gap during crises. Solving this in Gravataí means creating a model that communicates with Bangladesh, Guatemala, the Philippines, and so many other places where the climate is changing faster than the structures can keep up.

When nature accelerates, communication cannot be left behind.

Is the problem clearly explained and meaningful?

The problem is articulated with clarity, urgency, and depth. The framing of “collapse of communication” as a human failure rather than a technical one is compelling and meaningful. The student explains causes, symptoms, and consequences in a way that is emotionally resonant and analytically sound.

9 months ago

What is your innovative idea to solve the problem, and how will it work?

GSSA (Gravataí Support & Shelter Assist) emerged not from theory, but from urgency. Witnessing Brazil's most catastrophic environmental disaster, I realized such chaos was not inevitable—it was solvable. I propose a web-based platform transforming climate crisis response from disconnected to coordinated, creating a geographically organized hub connecting people in need, volunteers, and NGOs seeking to reach more people or receive support. It's not about inventing new technology, but using what people already have—smartphones, internet browsers, and the profound human instinct to help one another—and structuring that into something functional when everything else collapses.

GSSA's heart is an interactive crisis map (currently focused on Gravataí) displaying the user's location and nearby NGO markers. Clicking markers reveals critical information: capacity, target audience, services provided, and calculated distance. A single button opens Google Maps with directions. This eliminates desperate, inefficient searches for help—transforming "where can I go?" into instant, visual answers. No more walking kilometers in the rain to find overcrowded shelters. No more knocking on doors hoping someone knows something. In climate crises, no one expects displacement. People who never imagined needing shelter find themselves lost when nature abruptly forces them from home. The information now exists, centralized and accessible.

This bridge system between those in need and institutions allows contact via WhatsApp or email for assistance. However, GSSA aims beyond emergency contact—it's a true communication and support

network. I developed a volunteer registration system where individuals sign up for NGOs, listing skills (medicine, cooking, education, etc.), location, and availability. Volunteers navigate the map, discover nearby organizations, and filter opportunities matching their experience. When finding suitable opportunities, they apply digitally. Organizations review applications in administrative panels, viewing verified skills and contact information, then approve or decline. Approved volunteers receive confirmation and track commitments on personal dashboards. This transforms volunteer coordination from people appearing randomly—creating chaos rather than alleviating it—into strategic matching of skills to needs.

GSSA provides organizations an administrative panel for registration, verifying legitimacy to prevent fraud. Organizations pinpoint exact locations and describe their services. Real-time updates show services requested during crises, resources raised, and current capacity, ensuring families don't travel unnecessarily to overcrowded locations. This structured communication reaches thousands immediately, replacing chaotic WhatsApp groups and outdated Facebook posts with organized, verified information.

GSSA operates on three access levels, balancing accessibility with accountability. Unauthenticated visitors view maps and organization information without barriers—anyone in crisis accesses support immediately. Registered volunteers gain application and tracking abilities. Registered organizations post opportunities, manage volunteers, and update information. Technically, GSSA is built for crisis conditions: web-based architecture works on basic smartphones without app downloads or significant storage; the platform functions with intermittent internet connectivity; the interface uses large buttons, clear icons, and minimal text for users with varying digital literacy and stressed mental states.

Beyond emergencies, GSSA operates year-round. Organizations recruit volunteers for ongoing programs and contact other institutions to develop assistance actions—strengthening the social fabric that makes communities genuinely resilient even without imminent crises.

GSSA doesn't prevent floods. But it prevents the silence that follows them.

How strong and relevant is the core idea?

Excellent idea!

9 months ago

The core idea is strong, relevant, and clearly solves the stated problem. It directly addresses communication gaps during crises and translates empathy into actionable system design. The explanation is coherent, detailed, and grounded in real world constraints.

9 months ago

with global warming and these unexpected natural disasters happening there is a need for services like these now more than ever

9 months ago

Does the idea clearly address the selected challenge and SDG?

There is strong and direct alignment with climate resilience, community infrastructure, and disaster response themes. The solution clearly addresses the challenge focus and aligns with the spirit of relevant SDGs, even without over reliance on labeling.

9 months ago

What makes your innovation different or better than existing solutions?

GSSA’s uniqueness lies not in inventing a whole new technology but in innovatively combining existing tools to address a gap that sophisticated alternatives overlook.

Consider Google Maps, the world’s most sophisticated mapping platform. It effectively shows locations of organizations but not their real-time capacity, services, or whether they’re accepting people during an emergency. During the 2024 floods, or, taking another global example, the raging wildfires in Spain this year, a family could see a church's precise coordinates while having no idea whether it was operating as a shelter, whether it had available space, or what services it offered. GSSA adds the critical layer Google Maps can't: dynamic, emergency-specific information updated in real time by the organizations themselves. "This shelter has 15 beds available NOW, offers hot meals until 8pm, and welcomes families with children." That’s the difference between a static directory and a living, breathing crisis response system that prevents wasted journeys when every minute matters.

Furthermore, social media platforms such as WhatsApp and Facebook theoretically enable rapid information sharing, and during the floods, thousands, indeed, used them attempting to coordinate help. I personally joined about seven different WhatsApp groups. Messages arrived at overwhelming speed – hundreds per hour –with frequent misinformation, no actual organization, and information that became outdated within minutes yet remained visible indefinitely. Someone would just post "Shelter X has space" at 10am; by 11am it was full, but the message stayed, sending desperate families on futile journeys.GSSA solves this mainly by centralizing information in one verified platform, structuring it geographically rather than chronologically, and enabling instant updates directly from sources. That's the difference between a chaotic crowd shouting conflicting directions and an organized dispatch system with verified, current information.

Taking another optic, government emergency alert systems broadcast warnings effectively ("flooding expected in your area”) but rarely provide actual actionable solutions. They don't say "here are the five nearest shelters with real-time capacity and directions." They also rarely integrate grassroots organizations. Moreover, churches, NGOs, and community centers that actually provide the bulk of ground-level support in peripheral areas where governmental infrastructure is weakest. GSSA doesn't replace governmental systems. Not at all. Actually, my main focus now is to correlate this project with local governments. It complements them by focusing specifically on connecting people to local organizations, bridging the gap between official alerts and community-level response.

> Most fundamentally, existing solutions serve single user types: apps for finding services OR managing volunteers OR organizational coordination. GSSA uniquely serves all three simultaneously—displaced individuals, volunteers, and organizations—because effective crisis response requires coordinated, multi-directional information flow. A platform helping people find shelters but not helping those shelters organize volunteers providing assistance is incomplete.
>
> Finally, GSSA is designed from inception for replication. It's hyperlocal—built for Gravataí's geography and organizations—but architecturally universal. The same codebase deploys to any city by simply changing map coordinates and initial databases. This "clone and customize" model enables horizontal spreading across communities far faster than one-size-fits-all solutions requiring extensive customization. GSSA doesn't compete with existing tools. It orchestrates them.

Does the idea clearly stand out from existing solutions?

great personal examples help demonstrate how the product differs from existing platforms

9 months ago

The differentiation is clear and compelling. The student demonstrates strong competitive awareness, comparing their solution thoughtfully against Google Maps, social media platforms, and government alerts. The articulation of why existing tools fail in crisis conditions and how GSSA orchestrates rather than replaces them is particularly strong.

9 months ago

Upload your logo representing your innovation!

gssa_logo_transp.png (88 KiB download)

What type of group or funder would best fit to support your idea?

> Grassroot Organization/ Community Localization

How would your solution remain sustainable?

Sustainability isn't an afterthought for GSSA—it's embedded in its architecture. I've observed so many well-intentioned technological solutions collapse not because they failed to work, but because they failed to endure. Our question isn't whether GSSA can work during a single crisis---it's whether it can ultimately remain functional across years, multiple crises, and shifts in funding and attention.

Firstly, GSSA's technical sustainability begins with its deliberate simplicity. GSSA runs on Firebase, Google's cloud platform offering exceptionally generous free tiers—thousands of users can access the platform at zero cost. Even when expenses eventually emerge, they remain minimal, even under heavy usage. Besides, this web-based structure using universally documented technologies cuts out any chances of app-store fees, update distribution complexities, or even device compatibility problems. Once deployed, the platform mainly self-maintains, requiring updates only when organizations join, modify information, or during actual emergencies. This isn't accidental; it's designed resilience.

Furthermore, financial sustainability follows a sort of a three-stage evolution. Currently, development is self-funded—my time, minimal hosting costs. Thinking about scaling beyond Gravataí, I seek firstly to pursue grant funding from foundations focused on climate resilience, technology for social good, or civic innovation. A single grant could fund enhanced features—early warning system integration, multi-language support—and deployment to five to ten additional cities. Subsequently, I deeply envision municipal partnerships where city governments contribute modest annual fees, not as software payment—the platform remains free and open-source—but as contracts for localization: adding their organizations, training civil defense teams, customizing alert systems to municipal protocols. Cities already spend vastly more on less effective emergency communication infrastructure; GSSA offers superior outcomes at fractional cost. Long-term, this evolves into hybrid funding combining municipal partnerships, ongoing innovation grants, and potential corporate social responsibility sponsorships from technology companies demonstrating climate action commitments. The platform might generate modest revenue from optional premium features for larger NGOs—advanced analytics, volunteer management tools—while keeping all core crisis response features permanently free.

Most critically, sustainability depends on community ownership. GSSA isn't an external service imposed on communities—it's built with and for them. Local NGOs aren't customers; they're co-creators shaping how the platform evolves based on real-world needs. This creates intrinsic motivation for adoption and maintenance. In Gravataí, I'm establishing partnerships with anchor organizations that will champion the platform within their networks. These organizations possess deep community trust and perceive GSSA as enhancing, not replacing, their existing work. As the platform proves valuable, it becomes embedded in local emergency response protocols, referenced by civil defense, taught in community preparedness workshops, shared through word-of-mouth.

In the end, I want to make GSSA available as open-source software so that any community around the world can set up their own instance and add enhancements to the main codebase. As a result, a federation of locally run but technically interoperable platforms is created; in this dispersed approach, Rio's version may include capabilities that São Paulo's does not, and vice versa, but they share fundamental design and benefit from each other's advances. Sustainability isn't merely an outcome of GSSA. It's what GSSA is.

Does the student explain how the idea could realistically continue?



Excellent thought process here, with government grants and municipalities as funding sources.

9 months ago

The sustainability plan is realistic, staged, and well reasoned. The student addresses technical, financial, and community sustainability with concrete mechanisms such as grants, municipal partnerships, open source development, and low cost infrastructure. This shows unusually mature thinking for this level.

9 months ago

what happens if there is no internet access or power? how would an individual or family be able to access the service if they don't have access to the program?

9 months ago

How did working on this challenge change how you see the world or how you approach problems?

> I already had a background of extensive contact with NGOs and local communities, assisting them in their efforts to help others, especially through the creation of a student initiative (called Atrián) for the development of systems and automation of processes in local communities – such as the Casa dos Sonhos Foundation, AICA Gravataí, SABEVE, and many others. I have always sought to address and distance myself from the invisibility that local support faces in society. I want more and more people to understand its causes, how to help, and how to protect the most vulnerable. Because that is what I believe to be our reason for living: to serve. To help. To collaborate. To make a real impact, and to truly change lives, lives that can only be changed by people willing to live the extraordinary. But this challenge brought me an excellent and vital moment of reflection on everything that I and everyone in the state of Rio Grande do Sul went through in 2024, and how the silence that permeated that emergency situation could have been avoided.

Does the reflection show meaningful learning or growth?

The reflection is deeply personal and thoughtful. It clearly shows growth in perspective, motivation, and problem solving approach. The student connects past experience, the challenge process, and future intent in a cohesive and authentic way.

9 months ago

What is one thing you would explore more in your idea if given more time?

The most important thing for the future of GSSA was preparing an MVP and structuring the idea in such a short time. Currently, the prototype already meets much of its true purpose: connecting people in need with NGOs and volunteers. The relationship between people in vulnerability and community centers is already very well established, but I want to enhance the sector of the NGO-volunteer relationship. I aim to implement the functionality for the organization to publish projects, actions, and events for users, so they can follow the expressions of aid and see how they can help in each one. Furthermore, I intend to add the real-time pop-up mechanism, serving as an interactive resource map that clearly shows the user where to find what they need in times of crisis. Furthermore, I want to link the GSSA with a climate alert system, showing users on the platform itself when there is a risk of heavy rains or severe winds, contributing to the purpose of being a preparation and support channel/hub. Finally, I intend to set up a crisis preparation center for climate emergencies, detailing what to do, whom to assist, how to request help, how to prevent such situations, etc. I am still thinking about how I will introduce this dynamic, whether thru videos, texts, and relevant information, or if I will be able to install a specific chat-bot.
Over time, I will be adding to and refining this project, especially through partnerships and contacts with aid organizations, willing individuals, and municipalities. Scalability will take time, but the small spark I intend to ignite will already show what we can do and how we can react to future climate crises. We cannot fail to protect ourselves from catastrophes of such magnitude.

Why is contributing to this challenge important to YOU?

As I mentioned before, the NFTE World Series of Innovation gave me, above all, a moment to contemplate and reflect on the scars that climate crises leave on communities. It's something almost uncontrollable and unpredictable. And, above all, because I experienced firsthand what such a catastrophe means. This challenge sparked a desire to analyze my own reality and see what I could truly do to improve this scenario... to better prepare ourselves for another May 2024. Perhaps I thought, before, that I wouldn't be able to handle such a large demand. But as a Brazilian saying goes, "only those who dabble take risks." I used my technological knowledge from my 1200-hour Technical Course in Systems Development and challenged myself to research all the technologies and connections I would need to build a system like this. The mere possibility of validating my attempt to address this problem that has affected my home and so many other countries around the world gives me hope that I can truly make a difference for people who suddenly become homeless due to rain, wind, and snow.

Does the student explain why the idea matters and who it benefits?

The student clearly and convincingly explains why the idea matters, grounding it in lived experience from the 2024 floods and highlighting the real human consequences of communication failure during climate emergencies. The beneficiaries are clearly identified, displaced individuals, local NGOs, and volunteers, and the impact on dignity, safety, and coordination is articulated with depth and empathy.

9 months ago

Is the name of the idea and/or logo memorable and marketable? (Attachments shown below)

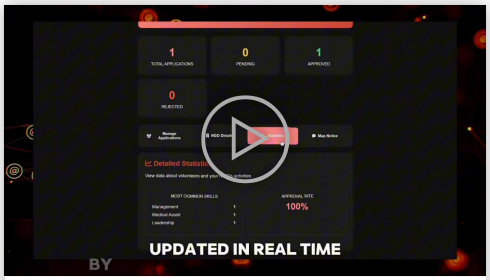
Would consider if you use Gravatai in the name- or make it more general as you hope to roll it out to various jurisdictions over time.

9 months ago

The name GSSA is meaningful and clearly tied to the local context, which strengthens authenticity and trust, especially for community based adoption. While the acronym is functional and professional, the student could further explore how the name might immediately convey purpose to someone unfamiliar with the project. The logo supports the idea well and feels appropriate for a civic and humanitarian platform.

9 months ago

Attachments



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(130.7 MiB download)

Comments on your entry

This is a very strong and thoughtful submission rooted in firsthand experience,

empathy, and a clear understanding of real world challenges during climate crises. The problem is well defined, the solution is practical and well reasoned, and the differentiation from existing tools is clearly articulated. The sustainability plan shows maturity and long term thinking, and the reflection demonstrates meaningful personal growth. As a next step, the student could further simplify how the idea is explained to first time users and consider how branding could instantly communicate the platform's purpose during high stress moments. Overall, this entry reflects advanced critical thinking, strong social awareness, and a genuine commitment to impact.

9 months ago

Overall I think that this is a great idea. I think it is definitely beneficial to those living in areas in the world or regions that have a lot of damaging climates especially when the storms or fires come on so suddenly and evacuations are necessary

9 months ago

great idea !

great presentation !

9 months ago

Your submission is truly outstanding. You demonstrate an exceptional understanding of the communities affected by climate-related disasters, going far beyond surface-level descriptions. Your ability to combine personal experience, data, and empathy creates an incredibly powerful narrative that shows real insight into the lives of the people you aim to support.

You clearly identify the users your solution serves, explaining their needs with precision and emotional depth. The problem you chose to address, communication collapse during climate emergencies, is described with clarity, urgency, and meaningful context. It is evident that you deeply understand why this issue matters and how it affects individuals, organizations, and volunteers.

Your idea, GSSA, stands out as innovative and relevant. You offer a well-structured, thoughtful, and realistic solution that fills a critical gap left by existing tools. The platform's design is smart, accessible, and grounded in real-world crisis experiences. Your differentiation from current solutions is clear, compelling, and technically sound.

You also present a very strong sustainability strategy. Your plan for long-term technical, financial, and community viability is well-reasoned and realistic, showing maturity beyond your age. The vision of open-source expansion and community ownership is especially inspiring.

Your personal reflection is thoughtful and heartfelt. It shows meaningful growth, a deep sense of purpose, and a genuine desire to serve others. You communicate clearly why this challenge matters to you and why your idea can make a difference.

Overall, your work is insightful, compassionate, technically strong, and truly impactful. You should be very proud of what you created—your voice, your vision, and your commitment shine through every part of the submission.

Congratulations!!

9 months ago